SMM Tool · Developer Specification

Campaign Learning Tab & Benchmarks

Per-campaign learning, benchmark setting (manual or best-performance), change tracking, KPI alerts and AI insights — for **Meta Ads** and **Google Ads**

Digi Carotene · Team Portal → Growth & Analytics · Version 1.0 · November 2026

Companion to: *Campaign Intelligence Roadmap (Meta Ads + Google Ads)*

What we're building. Every Meta and Google campaign gets a **Learning** tab. The tab holds a **benchmark** — either typed in by the team or calculated automatically from the campaign's own best-performing period — and shows how the campaign is doing against it, what was different when it performed best, what the team has changed, what worked, and what to try next. The paid marketing executive works from this tab every day.

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1. Purpose, users and scope

1.1 Problem

- Campaign performance is judged by feel — there is no agreed “good” number per campaign to work towards.
- Changes made on Meta / Google are not recorded with a reason, so nobody can say later what worked.
- Drops are noticed late, and the same mistakes repeat across clients and executives.

1.2 Users

Role	Main use of the Learning tab
Paid marketing executive	Daily: check gap to benchmark, annotate changes, act on recommendations, record learnings
Team lead	Approve benchmarks, review learnings, monitor executives' follow-through
Account manager	Use benchmark and learnings in client conversations and reports
Admin	Configure defaults (thresholds, window lengths), manage connections

1.3 Scope

In v1	Later (not v1)
Learning tab for every Meta Ads and Google Ads campaign	Learning tab at ad set / ad group level
Benchmark: manual, auto (best performance), client target	Industry benchmarks from external sources
Benchmark history, approval, “new best” detection	Client-facing view of the Learning tab
Gap-to-benchmark cards and chart with change markers	Auto-applying recommendations to Meta / Google
“What was different in the best period” comparison	Cross-client playbook search (portfolio-level)
Learnings feed (from impact analysis + AI) and team notes	
Recommendations with Accept / Reject / Modify	
Changes inbox and alerts that use the benchmark	

Dependency: this spec assumes Phase 1 of the roadmap is in place — daily Meta Ads and Google Ads metric sync into shared tables, connection health and “last synced” times. The Learning tab must show a clear stale-data warning if the campaign has not synced in the last 24 hours.

2. Glossary

Term	Meaning
Primary KPI	The one KPI the campaign is judged on, chosen from its goal: e.g. cost per lead, cost per WhatsApp conversation, CPA, ROAS, CPM. Set automatically from the Meta objective / Google campaign type; editable.
Secondary KPIs	3–5 supporting KPIs shown next to the primary (e.g. CTR, conversion rate, frequency, impression share).
Benchmark	The target value for the primary and secondary KPIs. One active benchmark per campaign at a time; older ones kept as history.
Benchmark mode	**Manual** (typed by the team), **Auto** (calculated from the campaign's best period), or **Client target** (agreed with the client; manual with a source label).
Best period	For auto mode: the date window(s) in the campaign's own history that produced the best primary KPI, after exclusions (section 7).
Gap	How far current performance is from the benchmark, as a percentage, signed so that positive = better than benchmark.
Status	**On track** / **Below** / **Off track**, from the gap (section 7.5).
Learning mode	The tab's state: **Collecting data** → **Tracking**; plus a **New best** flag when the campaign beats its benchmark.
Learning	A recorded insight about this campaign — what worked, what didn't, a hypothesis — from impact analysis, AI or a person.
Change	Any edit detected on Meta (activity log) or Google (change_event), optionally annotated by the executive with a reason.
Learning phase (platform)	Meta's / Google's own re-optimisation period after a significant edit. Not the same as “Learning mode” — the UI must label them differently: platform learning is shown as a grey *“In platform learning”* chip.

3. Navigation and where things live

Inside the existing Team Portal, under **Growth & Analytics**:

Route	Page	Notes
`/growth-and-analytics/campaigns`	Campaign list	All Meta + Google campaigns; filters: client, platform, status, learning mode, gap status
`/growth-and-analytics/campaigns/:id`	Campaign detail	Default tab = Overview
`/growth-and-analytics/campaigns/:id/learning`	Learning tab	Deep-linkable; this spec
`/growth-and-analytics/campaigns/:id/changes`	Changes tab	Change log for this campaign
`/growth-and-analytics/campaigns/:id/alerts`	Alerts tab	Alerts for this campaign
`/growth-and-analytics/campaigns/:id/structure`	Ad sets / Ad groups & Ads	Drill-down tables
`/growth-and-analytics/changes-inbox`	Changes inbox	Daily review across all campaigns
`/growth-and-analytics/alerts`	Alerts centre	Across all campaigns

Campaign detail tabs, in this order: **Overview · Learning · Changes · Alerts · Ad sets / Ad groups · Settings**. What each tab contains is specified in **section 8B** (summary table there). The Learning tab shows a small dot badge when there are unread learnings, pending recommendations or a “New best” flag.

Sidebar: add *Campaigns*, *Changes inbox* (with count of changes awaiting annotation) and *Alerts* under Growth & Analytics, below Campaign Analytics.

4. Learning tab — UI specification

Wireframes of the tab in its normal (**Tracking**) state for a **Meta Ads** campaign (Figure 1a) and a **Google Ads** campaign (Figure 1b), plus the Google-only panels (Figure 1c). Numbers in orange circles map to the section descriptions in 4.1, which apply to both platforms.

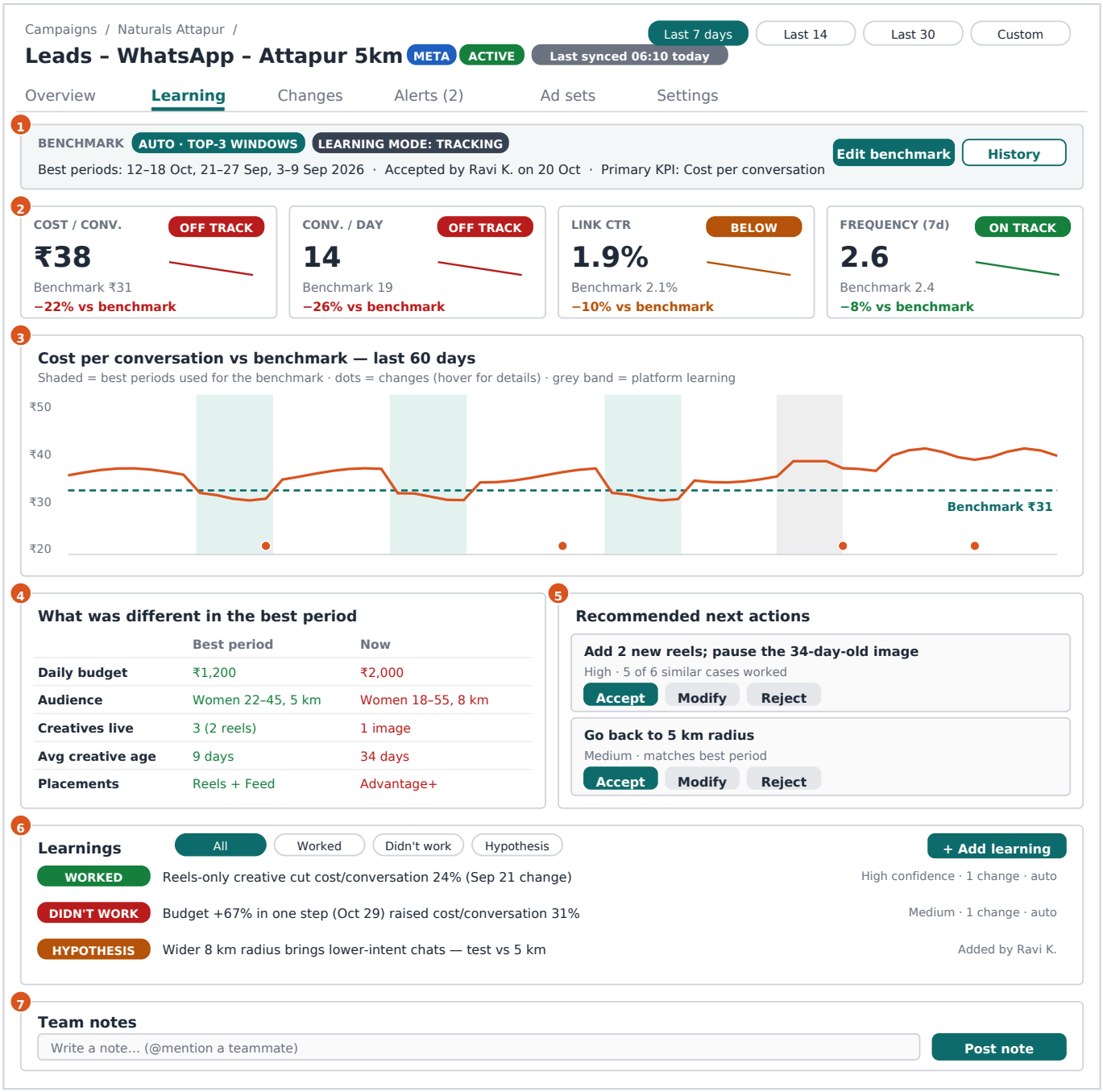


Figure 1a — Learning tab, Meta Ads campaign (Click-to-WhatsApp example), Tracking state. Not to scale; follow the existing portal design system for colours, spacing and typography.

4.0 Google Ads version of the Learning tab

The Google Ads Learning tab uses **the same layout and the same components** as the Meta tab — build it once and feed it platform-specific data. What changes: the platform chip, the child tab name (**Ad groups** instead of Ad sets), the default KPI cards, the rows in “What was different in the best period”, the source of the grey platform-learning bands, and an extra row of **Google-only panels** (Figure 1c) placed between the chart ③ and the best-period comparison ④.

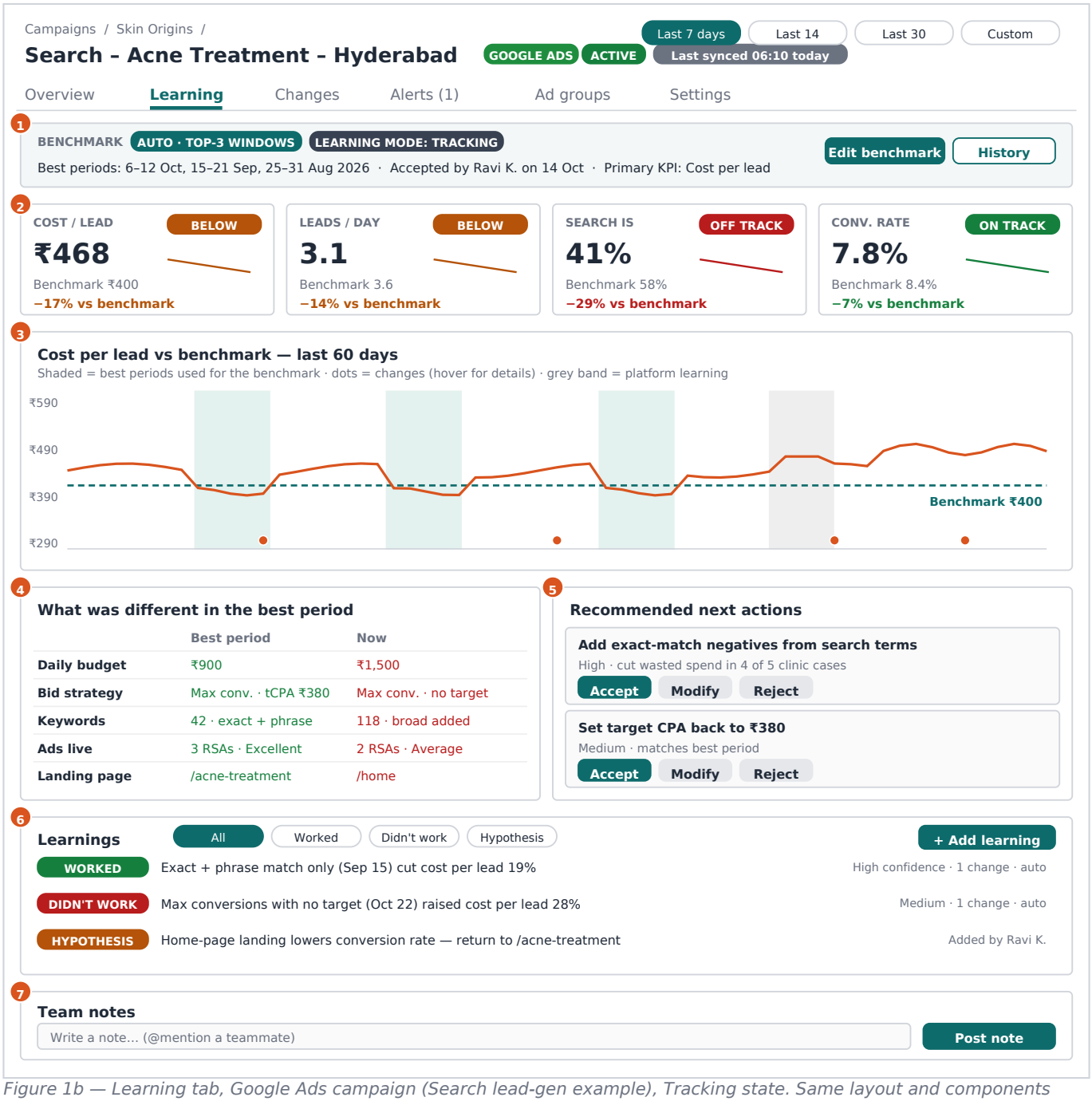


Figure 1b — Learning tab, Google Ads campaign (Search lead-gen example), Tracking state. Same layout and components as Meta; only data, labels and the extra panels differ.

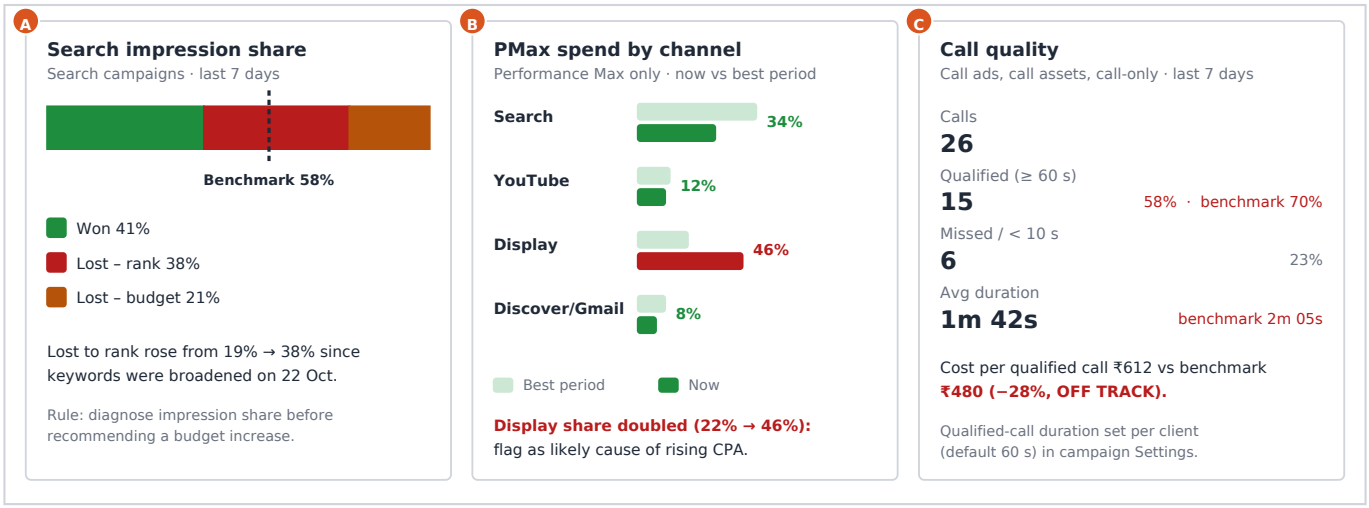


Figure 1c — Google-only panels, shown in a row between the chart (③) and the best-period comparison (④). Each panel appears only when it applies to the campaign type.

Part of the tab	Meta Ads	Google Ads
Platform chip / child tab	META (blue) · **Ad sets**	GOOGLE ADS (green) · **Ad groups** (PMax: **Asset groups**)
Default KPI cards ②	From the campaign objective (section 7.1) — e.g. cost per conversation, conversations/day, link CTR, frequency	From the campaign type (section 7.1) — e.g. cost per lead, leads/day, search impression share, conversion rate; ROAS for PMax / Shopping; cost per qualified call for call campaigns
Grey “platform learning” bands ③	Ad set learning stage (learning / learning limited)	Bidding strategy status = learning
Extra panels	**Creative fatigue** (same slot): per creative — age, frequency, CTR trend, hook rate; flags creatives older than 21 days with falling CTR	**A** Impression share (Search) · **B** PMax channel split (PMax) · **C** Call quality (campaigns with calls). Show only those that apply
Best-period diff rows ④	Budget, bid strategy, audience, creatives live, creative age, placements, frequency	Budget, bid strategy + target CPA/ROAS, keywords (count, match types), ads live + ad strength, landing page, impression share; PMax: asset groups, audience signals
Typical recommendations ⑤	Refresh creatives, narrow / widen audience, adjust radius, step budget in ≤ 20% moves	Add negatives, tighten match types, restore target CPA, fix ad strength, raise budget only when lost-to-budget is high
Change source (Changes tab)	Ad account activity log	change_event
Currency / attribution note	Store attribution window with the benchmark	Store conversion actions included in “Conversions” with the benchmark

4.1 Section-by-section behaviour

Header (above the tabs)

- Breadcrumb, campaign name, platform chip (Meta blue / Google green), campaign status chip, “Last synced” chip.
- If last sync is older than 24 h, replace the chip with a red banner across the tab: *“Data is stale — last synced 2 days ago. Numbers below may be out of date.”* with a link to Manage Accounts.
- Date-range pills control the **current** values on cards and the chart range. Default **Last 7 days**. The benchmark itself never changes with the date range.

① Benchmark strip

- Shows: mode chip (**Auto · method** / **Manual** / **Client target**), Learning mode chip, best-period dates (auto) or source note (manual), who set/approved it and when, primary KPI.

- Buttons: **Edit benchmark** (opens the drawer, section 5) and **History** (side panel listing previous benchmarks with dates, values, who, reason).
- If benchmark status is **Pending approval**, show an amber chip and, for team lead / admin, **Approve** / **Reject** buttons inline.
- If the campaign's objective / optimisation event / conversion action changed after the benchmark was set, show a red line: *“Campaign goal changed on 2 Nov — this benchmark may no longer apply.”* with **Recalculate**.

② KPI vs benchmark cards

- One card for the primary KPI (always first) plus up to 4 secondary KPIs (horizontal scroll on small screens).
- Each card: KPI label, current value for the selected range, benchmark value, signed gap %, status chip, 14-day sparkline.
- Gap colour and chip follow section 7.5. A KPI with too little data in the range shows “Not enough data (6 of 10 results)” instead of a gap.
- If the campaign is in platform learning, show a grey *“In platform learning”* chip on the cards and do not colour them red.
- Clicking a card switches the chart (③) to that KPI.

③ Performance vs benchmark chart

- Line: daily value of the selected KPI over the chosen range (default 60 days for this chart regardless of the pills, with its own toggle 30 / 60 / 90 / All).
- Dashed horizontal line: benchmark value. Shaded columns: the best-period windows (auto mode only). Grey bands: platform learning periods.
- Dots on the x-axis: changes from the change log. Hover → tooltip with change summary, who, reason (if annotated) and impact verdict; click → opens the change in the Changes tab.
- Optional toggle: 7-day rolling average (default ON for cost KPIs, to smooth daily noise).

④ What was different in the best period

- Auto mode only (hidden for manual with a hint “Available when the benchmark is based on the campaign's own best period”).
- Compares settings during the best period vs today, from daily settings snapshots: budget, bid strategy / target, audience or keywords summary, number and type of creatives live, average creative age, placements, frequency (Meta), impression share (Google), landing page.
- Highlight rows that differ. Rows: Meta → budget, bid strategy, audience, creatives, creative age, placements, frequency. Google → budget, bid strategy, target CPA/ROAS, keyword count / match types, ads live, impression share, landing page.

⑤ Recommended next actions

- Up to 3 open recommendations (AI or rule-based), each with title, confidence, evidence line and three buttons.
- **Accept** → creates a *planned change* pre-filled from the recommendation (it then appears in the Changes inbox for verification next day).
- **Modify** → same, but opens the planned-change form for editing first. **Reject** → asks for an optional reason (dropdown + text).
- Below the cards: “View all (5)” and “Why am I seeing this?” (shows the evidence JSON rendered as readable bullets).

⑥ Learnings feed

- Cards tagged **Worked** / **Didn't work** / **Hypothesis** / **Watch**; filter chips above.
- Sources: (a) automatic from change-impact verdicts, (b) AI weekly learnings, (c) added by a person via **+ Add learning**.
- Each card: text, linked change(s), confidence, source (auto / AI / person), date. Actions: **Pin**, **Add to client playbook** (later), **Edit** (person-added only), **Dismiss** (with reason).

- Pinned learnings stay at the top. Newest first otherwise. Paginate at 20.

⑦ Team notes

- Simple threaded notes with @mentions (notify via existing portal notifications). Notes are free text and are also passed to the AI as context.

5. Set Benchmark drawer — UI specification

Opens from **Edit benchmark** as a right-side drawer (about 480 px wide). Three modes; the preview table always shows what will be saved.

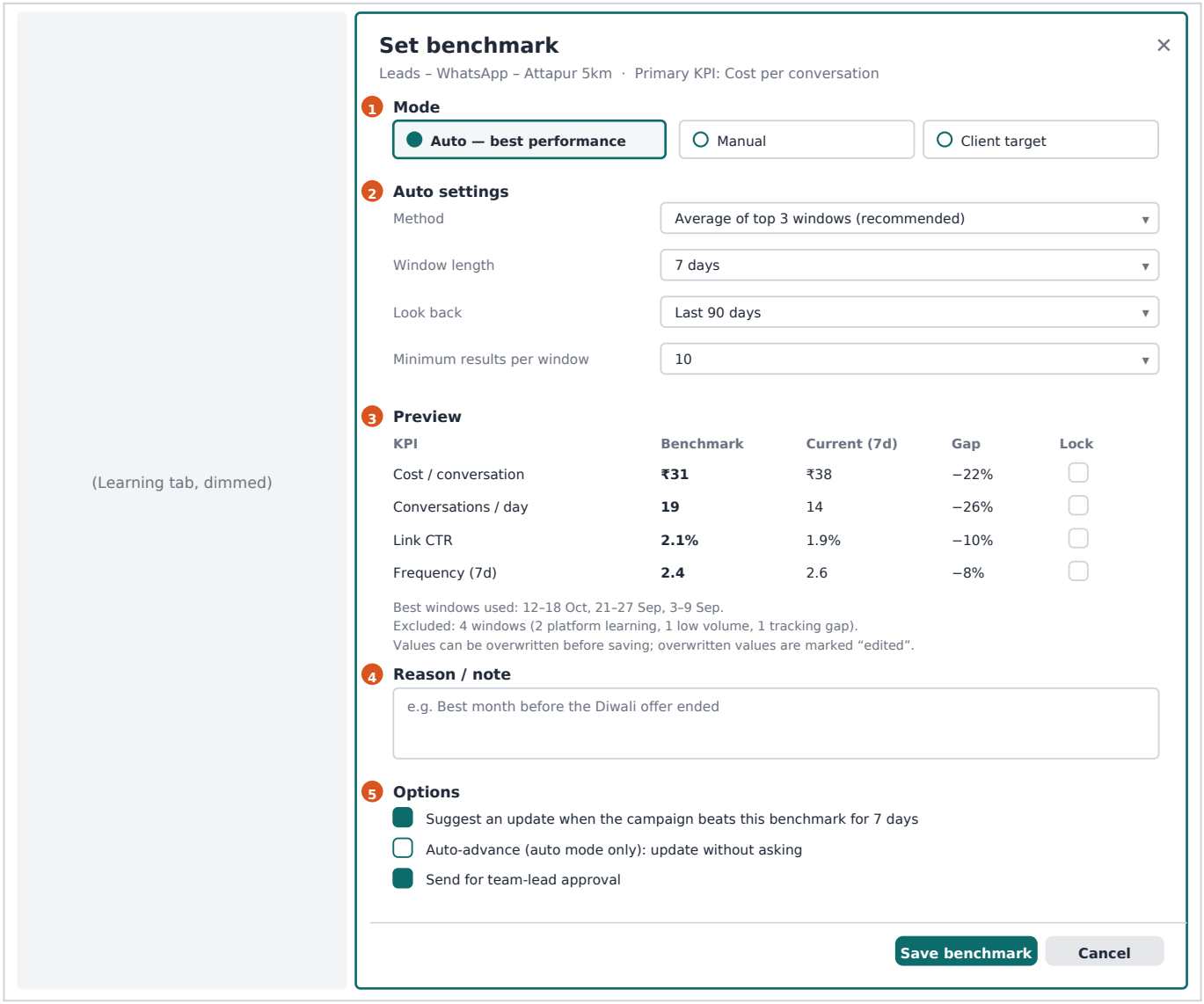


Figure 2 — Set Benchmark drawer, Auto mode. ① mode ② auto settings ③ preview ④ reason ⑤ options.

Mode	Fields shown	Validation
Auto — best performance	Method, window length (7 / 14), look-back (30 / 60 / 90 / 180 days / all), minimum results per window; preview with lock checkboxes; reason; options	Needs ≥ 21 days of history and ≥ 1 eligible window. If none: show “Not enough history for an automatic benchmark yet” with link to Manual.
Manual	Value input per KPI (primary required, secondaries optional), reason (required), options	Primary KPI value > 0; ROAS / CTR / rates within sensible ranges (e.g. CTR 0–100%).
Client target	Same as Manual + “Agreed with” (contact name) and “Agreed on” (date)	Same as Manual; reason optional.

- **Lock** (auto mode): a locked KPI keeps its current benchmark value when the auto benchmark is recalculated later.
- Saving creates a new row in `campaign_benchmarks` and marks the previous one *superseded*. Nothing is ever deleted.
- If “Send for team-lead approval” is ticked (or the user is an executive and the admin setting requires approval), the new benchmark is *pending*; the previous one stays active until approval.

6. Learning mode states

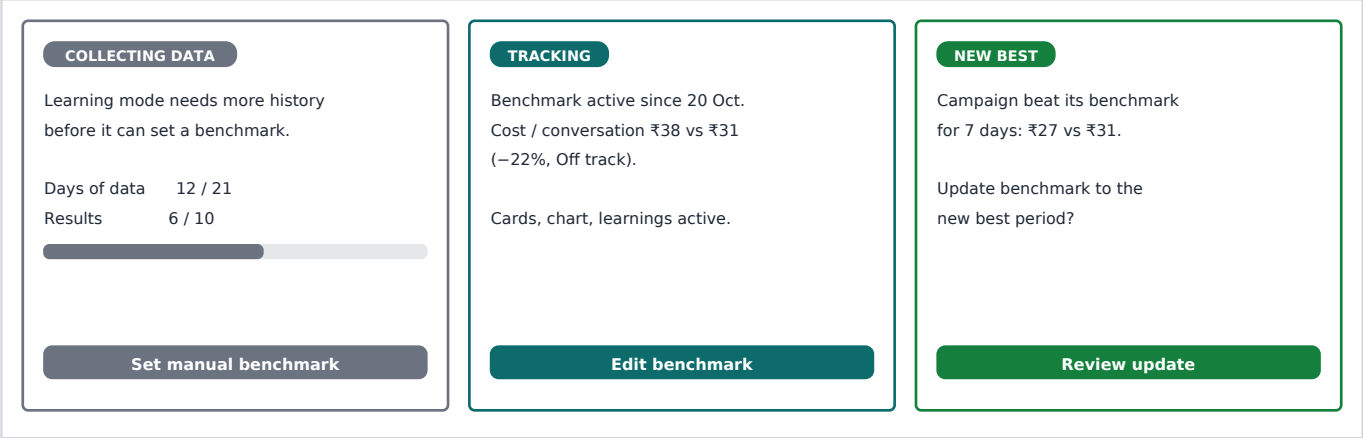


Figure 3 — Learning mode states shown in the benchmark strip area.

State	Enter when	UI	Exit when
Collecting data	Campaign has < 21 days of synced data **or** no eligible window (section 7.2) **and** no manual benchmark	Progress bars (days, results); cards show current values only, no gap; chart without benchmark line; learnings feed still active	Manual benchmark saved, or first auto benchmark calculated and accepted
Tracking	An active benchmark exists	Full tab as in Figures 1a / 1b	Benchmark deleted (admin only) or campaign archived
New best (flag on Tracking)	Primary KPI better than benchmark by ≥ 5% over the last 7 complete days **and** volume ≥ minimum **and** not in platform learning	Green banner in strip with **Review update** → opens drawer pre-filled with new auto values	User updates or dismisses (dismiss hides it for 14 days); auto-advance updates it silently and logs to history

7. Benchmark logic

7.1 Primary and secondary KPIs by default

Platform	Campaign goal	Primary KPI (direction)	Default secondary KPIs
Meta	Leads — Instant form	Cost per lead (lower)	Leads/day, CTR (link), form conversion rate, frequency
Meta	Leads — Click-to-WhatsApp / Messenger	Cost per conversation (lower)	Conversations/day, link CTR, frequency, CPM
Meta	Sales	ROAS (higher) or CPA (lower)	Purchases/day, CTR, add-to-cart rate, frequency
Meta	Traffic	Cost per landing-page view (lower)	LPV rate, link CTR, CPM
Meta	Awareness / Engagement	CPM (lower) or cost per ThruPlay (lower)	Reach, frequency, hook rate, hold rate
Google	Search — leads	Cost per conversion (lower)	Conversions/day, CTR, conv. rate, search impression share
Google	Performance Max	ROAS (higher) or CPA (lower)	Conversions/day, conv. value, Search share of spend
Google	Call / local	Cost per qualified call (lower)	Qualified calls/day, CTR, conv. rate
Google	Display / Demand Gen / Video	CPA (lower) or CPV / CPM (lower)	CTR, viewability / view rate

The user can change the primary KPI in campaign Settings; doing so marks the active benchmark as *needs review*.

7.2 Auto benchmark — algorithm

Inputs: daily metrics for the campaign (`ad_metrics_daily`), platform learning periods, data-quality flags, settings: window length L (default 7), look-back (default 90 days), minimum results per window R (default 10), method (default top-3 average).

1. **Build candidate windows.** Every rolling window of L complete days inside the look-back, ending no later than **3 days ago** (conversions are still being added for the most recent days).
2. **Drop ineligible windows.** Remove a window if any of these is true:
   – it overlaps a platform learning period by more than 2 days;
   – results in the window $< R$;
   – spend in the window $< 50\%$ of the campaign's median spend for a window of the same length (stops tiny-budget lucky weeks);
   – it contains a day flagged as a data problem (sync gap, tracking break: 0 conversions on a day where clicks $\geq 3\times$ the median daily clicks);
   – it falls before the campaign's last goal change (objective, optimisation event or primary conversion action changed).
3. **Score each window** on the primary KPI computed from the window's totals (e.g. total spend ÷ total results — never an average of daily ratios).
4. **Pick the best windows** according to the method:
   – **Top-3 average (default):** take the 3 best non-overlapping windows; benchmark = KPI from their combined totals;
   – **Single best window:** the one best window;
   – **80th percentile:** the value at the 80th percentile of eligible windows (best direction) — the most conservative.
5. **Secondary KPIs come from the same windows** (combined totals), so the benchmark is one real, consistent state of the campaign rather than the best of each metric picked from different weeks.
6. **Store** the values, the window dates, the method and settings, the count of eligible / excluded windows with reasons, and the platform.

Non-overlapping means: after picking the best window, remove all windows that overlap it, then pick the next best. If fewer than 3 eligible non-overlapping windows exist, use as many as there are and show “based on N window(s)” in the strip.

```
function computeAutoBenchmark(campaign, cfg = {L: 7, lookbackDays: 90, minResults: 10, method: 'top3'}) {
  const days    = getDaily(campaign.id, today - cfg.lookbackDays, today - 3)      // complete days only
  const windows = rollingWindows(days, cfg.L).filter(w =>
        learningOverlapDays(w) <= 2 && w.results >= cfg.minResults &&
        w.spend >= 0.5 * medianWindowSpend(days, cfg.L) && !w.hasDataIssue &&
        w.start > campaign.lastGoalChangeAt)
  if (windows.length === 0) return { status: 'insufficient_data', excluded: reasons }
  windows.sort(byPrimaryKpi(campaign.primaryKpi))                                    // best first
  const picked = cfg.method === 'p80' ? [percentileWindow(windows, 0.8)]
               : pickNonOverlapping(windows, cfg.method === 'top3' ? 3 : 1)
  const totals = sumTotals(picked)                                                    // spend, results, clicks...
  return { values: kpisFromTotals(totals, campaign.kpis), windows: picked.map(w => [w.start, w.end]),
           eligible: windows.length, excluded: reasons, method: cfg.method }
}
```

7.3 Manual and client-target benchmarks

- Values typed by the user are stored as-is. Reason is required for Manual.
- The system still computes the auto values in the background and shows them as a hint in the drawer (“Your best period was ₹31”) — but never overwrites a manual benchmark.
- If the campaign stays > 40% worse than a manual benchmark for 28 days, show a soft prompt: “This target hasn't been reached in 4 weeks — review it?”

7.4 Recalculation and history

- Auto benchmarks are **recalculated weekly** (Monday 04:00 IST). If the new values differ from the active ones by ≥ 5% on the primary KPI, create a *suggested* benchmark and raise the **New best** or **Benchmark update available** flag; apply automatically only when auto-advance is on.
- Locked KPIs keep their value on recalculation.
- All versions are kept: `status` = active · pending · suggested · superseded · rejected.

7.5 Gap and status

Direction-aware gap, so that a positive number always means better than benchmark:

```
lower-is-better KPIs (cost per result, CPA, CPC, CPM, frequency):  gap = (benchmark − current) / benchmark
higher-is-better KPIs (ROAS, CTR, conversion rate, results/day):     gap = (current − benchmark) / benchmark

status:  gap ≥ −10%            → ON TRACK  (green)
         −25% ≤ gap < −10%     → BELOW     (amber)
         gap < −25%            → OFF TRACK (red)
         volume < minimum      → NOT ENOUGH DATA (grey, no gap shown)
         in platform learning  → show gap, grey chip "In platform learning", never red
```

Thresholds (10% / 25%) are defaults stored in `client_kpi_targets` and editable per client by account managers. The alert engine (roadmap section 4) uses the same gap: OFF TRACK on the primary KPI for 3 consecutive complete days raises an **Act** alert; BELOW for 5 days raises a **Watch** alert.

7.6 Platform notes

- **Meta:** budgets may live at ad set level (ABO) — benchmark at campaign level, sum ad sets. Read the ad set learning stage for the “platform learning” bands. Store the attribution setting with the benchmark; if it changes, mark the benchmark *needs review*.
- **Google:** read the bidding-strategy status for learning bands. For call campaigns, “qualified call” = call over the client's minimum duration (default 60 s).
- **Currency:** benchmark stored in the ad account currency; display with the account's symbol.

8. Related screens — campaign list, change log, alerts

8.1 Campaign list

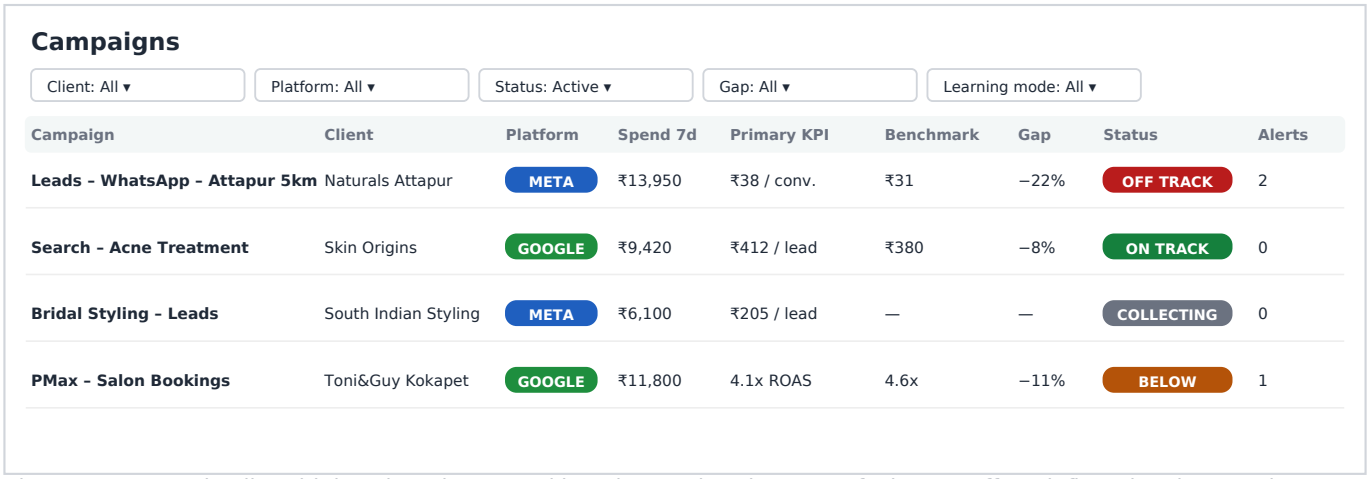


Figure 4 — Campaign list with benchmark, gap and learning-mode columns. Default sort: Off track first, then by spend.

8.2 Where campaign changes are recorded — overview

Every change to a Meta or Google campaign ends up in **one change log** (`campaign_changes` + `change_annotations`). It gets there in two ways, and the team sees it in three places:

How a change gets in	Who / what	When
Detected automatically	Change-capture job reads Meta's ad account activity log and Google's change_event	Every day at 03:00 IST, for the previous day
Logged in advance (planned)	Executive clicks **+ Log planned change** (Changes tab or inbox), or **Accept** on a recommendation	Any time, before making the change on Meta / Google

Where the team sees it	Purpose	Figure
Changes inbox (sidebar, with count badge)	Daily to-do: every detected change still missing a reason, across all clients and both platforms	6b
Campaign → Changes tab	Full history of one campaign: what changed, who, why, verification, measured impact	6a
Campaign → Learning tab	Changes as dots on the chart; impact verdicts become learnings	1a / 1b
Explain / Log change drawer	The form used from all of the above	6c

Daily routine for the paid marketing executive: (1) before changing anything on Meta / Google, log it as a planned change — or accept a recommendation; (2) make the change on the platform; (3) next morning, open the Changes inbox: planned changes now show **Verified / Mismatch / Not applied**, and any change made without logging shows as **Needs reason** — explain it there. Target: inbox at zero by 11:00 AM.

8.3 Campaign → Changes tab

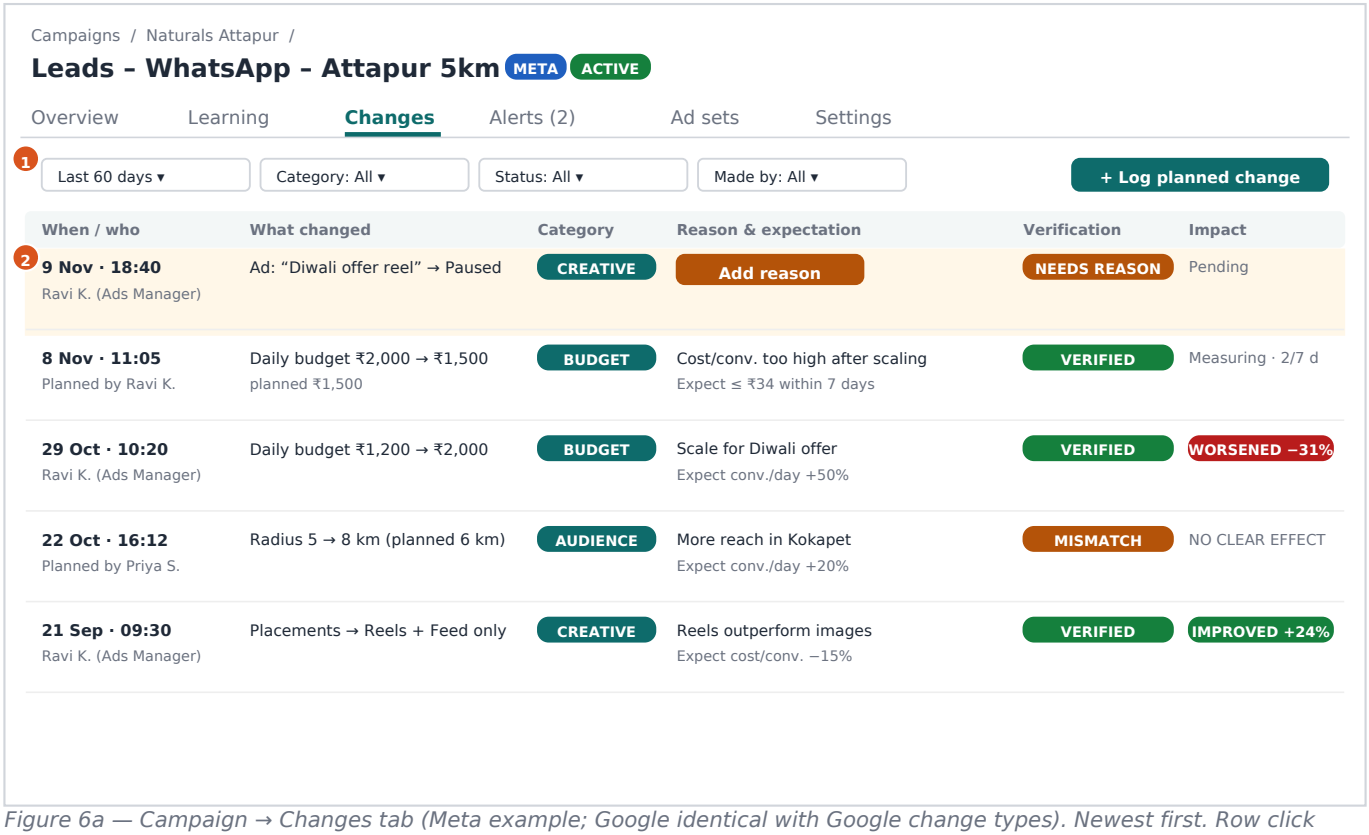


Figure 6a — Campaign → Changes tab (Meta example; Google identical with Google change types). Newest first. Row click expands full detail: old/new values, raw platform record, linked recommendation, before/after chart.

- One row per change. Source of truth = the platform record; the portal adds the reason, expectation, verification and impact.
- **Needs reason** rows (changes made without logging) are highlighted and float to the top until explained.
- **Verification**: Verified · Mismatch (shows planned vs actual) · Not applied (planned, not found on platform after 2 days) · Needs reason.
- **Impact**: “Measuring · x/7 d” until the after-window completes, then Improved / No clear effect / Worsened / Inconclusive (from section 5 of the roadmap; platform learning days are skipped).
- Change types captured — **Meta:** budget, bid strategy / cost cap, schedule, status, audience/targeting, placements, optimisation event, creative added/paused/edited. **Google:** budget, bid strategy / target CPA / ROAS, status, keywords added/removed/match type, negatives, ads added/paused/edited, assets, audience signals, locations, schedule, landing page.
- Minor system changes (e.g. automated ad review, auto-applied recommendations from Google) are captured but tagged *System* and hidden by default (filter to show).

8.4 Changes inbox (daily review across all campaigns)

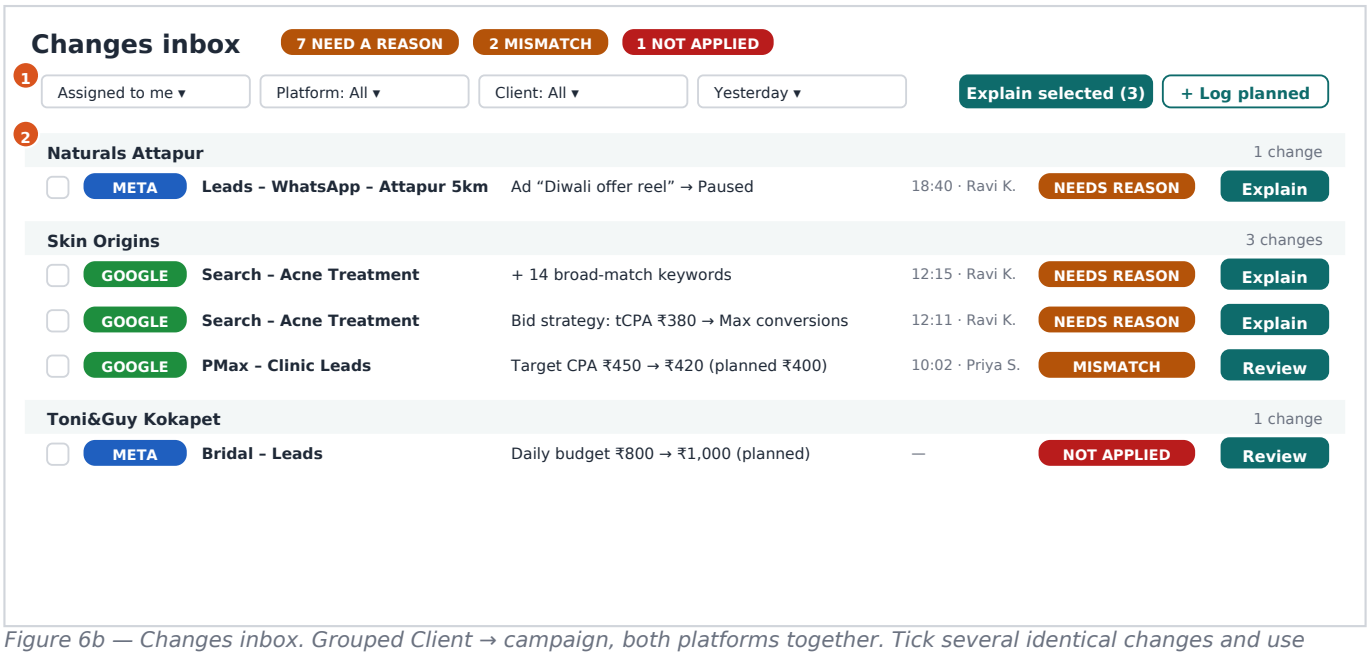


Figure 6b — Changes inbox. Grouped Client → campaign, both platforms together. Tick several identical changes and use “Explain selected” to give them one reason.

- Shows only items needing action: **Needs reason**, **Mismatch**, **Not applied**. Verified planned changes do not appear (they are already explained).
- Default filter **Assigned to me** (the executive's clients). Team lead sees all and a completion figure: “% of yesterday's changes explained by 11:00”.
- Sidebar badge = count of open items. Daily digest (09:30) links straight here.
- Mismatch → **Review** opens the drawer showing planned vs actual, with options: *Accept actual value* (updates the record) or *Note why it differs*.
- Not applied → **Review**: *Still planned* (keeps waiting, new date) · *Cancelled* (with reason) · *Applied differently* (pick the matching detected change).

8.5 Explain change / Log planned change drawer

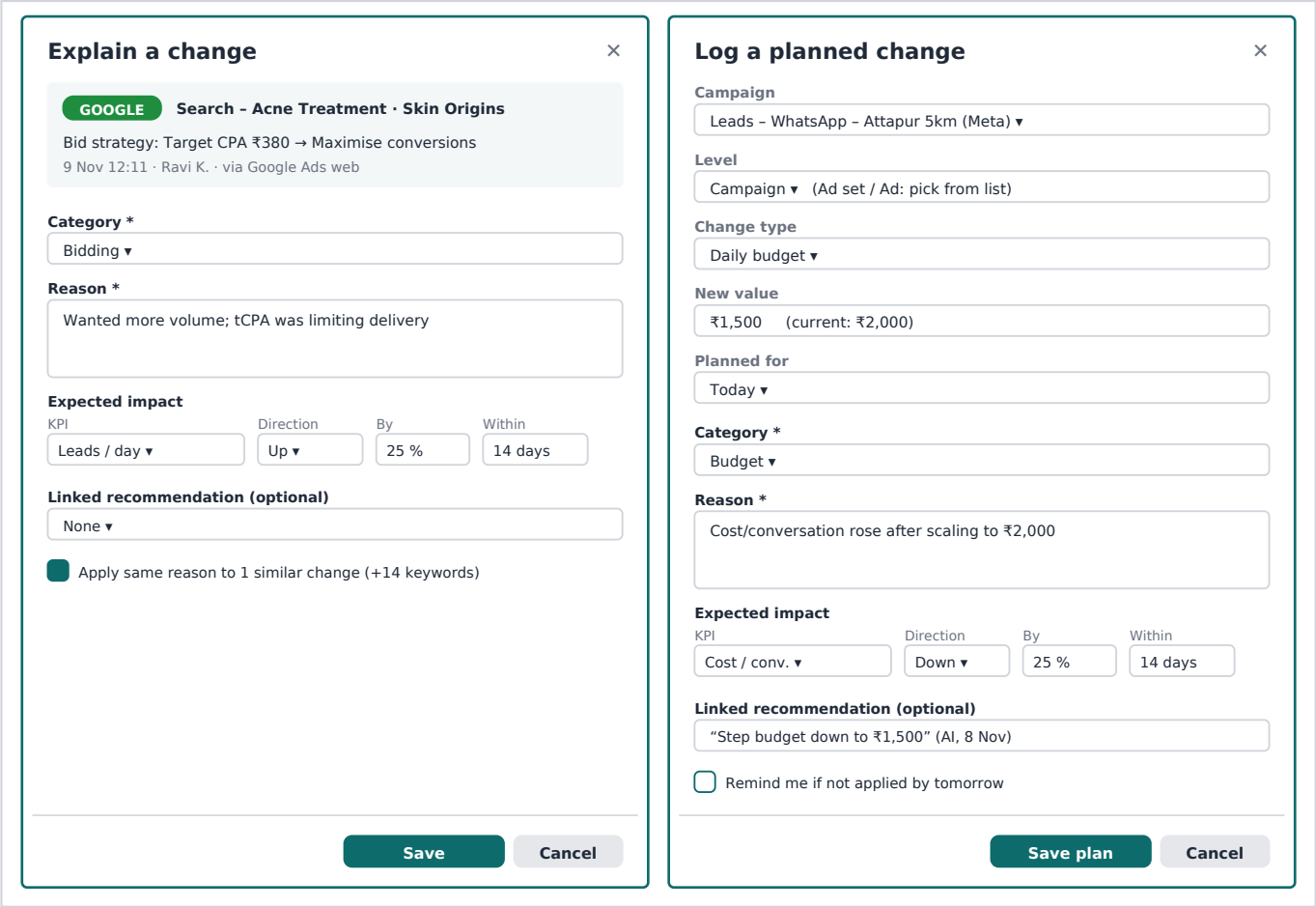


Figure 6c — Left: explaining a detected change (Google example). Right: logging a planned change before making it (Meta example). Both open from the Changes inbox, the Changes tab, chart dots on the Learning tab, and Accept / Modify on a recommendation.

Field	Rules
Category *	Budget · Bidding · Audience / targeting · Keywords · Creative / ads · Placement · Landing page · Tracking · Status · Other
Reason *	Free text, min 10 characters. Passed to the AI as context.
Expected impact	Optional but encouraged: KPI (campaign's KPIs), direction, size %, days. Used to judge "met expectation?" next to the impact verdict.
Planned change fields	Campaign, level (campaign / ad set or ad group / ad), change type (list per platform — section 8.3), new value (typed per change type), planned date.
Verification (planned)	Next change-capture run matches on campaign + level + change type within planned date + 2 days. Equal value → Verified; different value → Mismatch; nothing found → Not applied.
Bulk explain	Same reason saved to every selected change; each keeps its own row and impact.
Edit history	Reasons can be edited by the author or team lead; every edit kept (who, when).

8.6 Alerts centre and Alerts tab

- Alert row: severity chip (Act / Watch), campaign, KPI, current vs benchmark, gap, days in state, linked changes (last 14 days), **Acknowledge** / **Resolve** (with note).
- Hard-rule alerts from the roadmap (zero delivery, rejected / disapproved ads, payment problems, tracking stopped) appear here too, with no benchmark column.

8B. Campaign detail tabs — full contents

Every Meta and Google campaign opens on the same six tabs. This section lists exactly what each tab contains. Learning (section 4) and Changes (section 8.3) are specified above; the other four are below. Build each tab once and feed it platform data — the Meta and Google columns show what differs.

Tab	Purpose (one line)	Meta Ads specifics	Google Ads specifics	Spec
Overview	How is the campaign doing right now, and is it healthy?	Reach, frequency; breakdowns by age/gender, placement, region; top creatives	Impression share; breakdowns by device, location, hour, network; top keywords and search terms	8B.1
Learning	Benchmark, gap, what worked, what to do next	Creative fatigue panel	Impression share, PMax channels, call quality panels	4
Changes	Every change, why it was made, and its impact	From activity log	From change_event	8.3
Alerts	Problems on this campaign and how they were handled	Learning limited, rejected ads, payment, pixel / CAPI	Disapproved ads, limited by budget, conversion tag	8B.3
Ad sets / Ad groups	Drill-down below the campaign	Ad sets → Ads (creatives)	Ad groups → Ads → Keywords → Search terms; PMax: Asset groups	8B.2
Settings	How this campaign is judged and who owns it	Result = chosen action type; attribution window	Conversions = chosen conversion actions; qualified-call duration	8B.4

8B.1 Overview tab

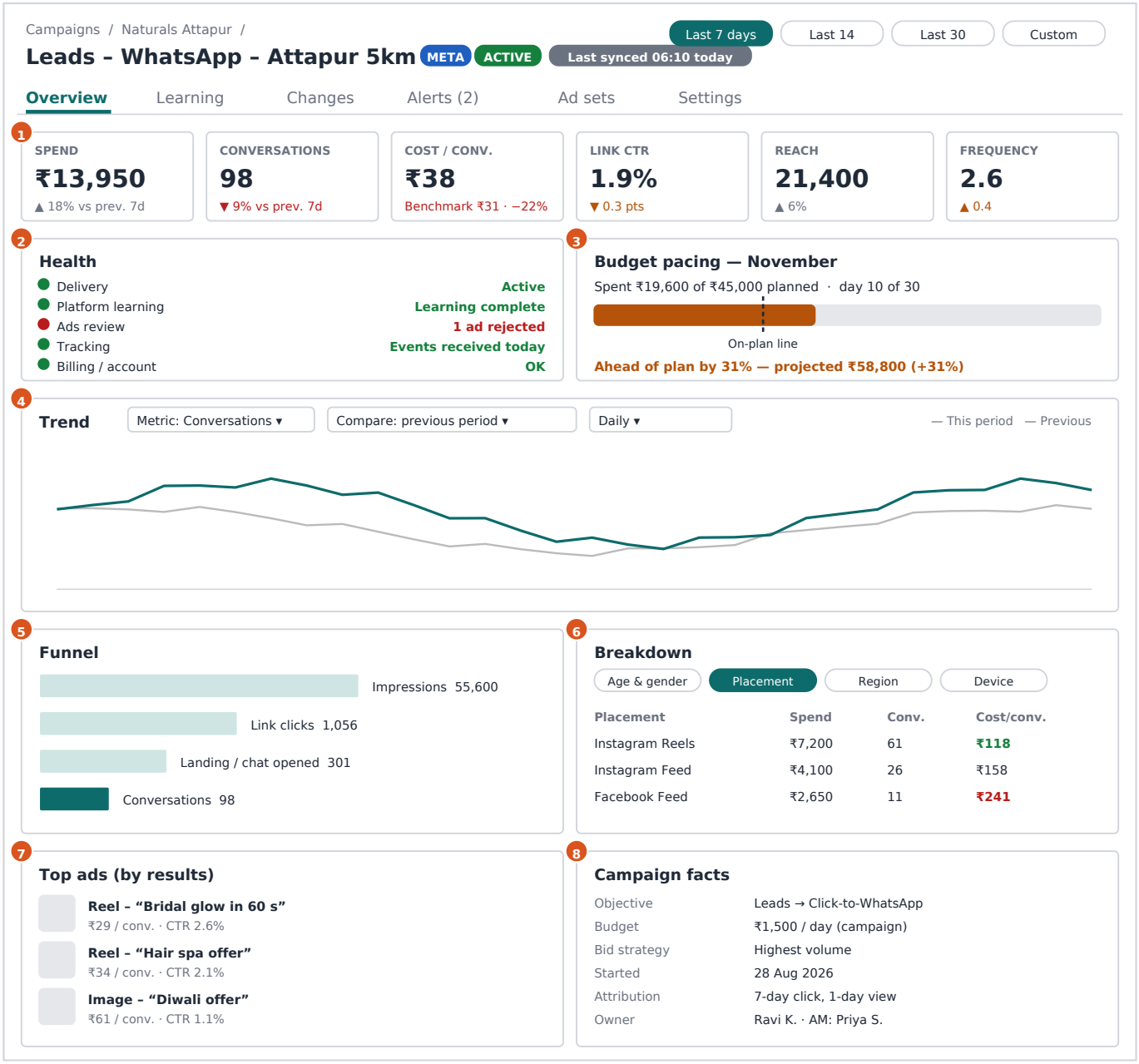


Figure 7 — Overview tab (Meta example). Google version: same blocks with Google KPIs and breakdowns (table below).

Block	Contents	Meta Ads	Google Ads
① **KPI row**	6 cards, value for the selected range + change vs previous period of equal length. The primary KPI card also shows the benchmark gap.	Spend, results (by objective), cost per result, link CTR, reach, frequency	Spend, conversions, cost / conversion (or ROAS), CTR, avg. CPC, search impression share (Search) / conv. value (PMax)
② **Health**	Traffic-light list; red items link to the Alerts tab	Delivery, ad set learning stage, ad review (rejected), pixel / CAPI events, account / payment status	Delivery, bid-strategy learning, policy (disapproved), conversion tag activity, billing, limited by budget
③ **Budget pacing**	Month spend vs planned budget (from client targets), on-plan line, projected month-end	Campaign or summed ad set budgets	Campaign budget / shared budget
④ **Trend chart**	Metric selector, compare previous period, daily / weekly; change markers as on the Learning tab	Any KPI in ①	Any KPI in ①

Block	Contents	Meta Ads	Google Ads
⑤ **Funnel**	Stage counts with drop-off %	Impressions → link clicks → landing views or chats opened → results	Impressions → clicks → conversions (→ qualified calls for call campaigns)
⑥ **Breakdown**	Switchable table, sorted by spend; best / worst cost per result highlighted	Age & gender, placement, region, device	Device, location, day & hour, network (Search / Partners / Display)
⑦ **Top items**	Top 3 by results with thumbnail, cost per result, CTR; “View all” → child tab	Ads (creatives)	Keywords (Search) · asset groups (PMax) · ads (Display / Video)
⑧ **Campaign facts**	Read-only settings pulled from the platform + owners from portal	Objective, budget, bid strategy, start / end, attribution, owners	Campaign type, budget, bid strategy + target, networks, locations, start / end, owners

8B.2 Ad sets / Ad groups tab

The drill-down below the campaign. Meta: **Ad sets → Ads**. Google: sub-tabs **Ad groups · Ads · Keywords · Search terms** (Search), or **Asset groups · Assets** (Performance Max). Wireframe shows the Google Search version, the most detailed.

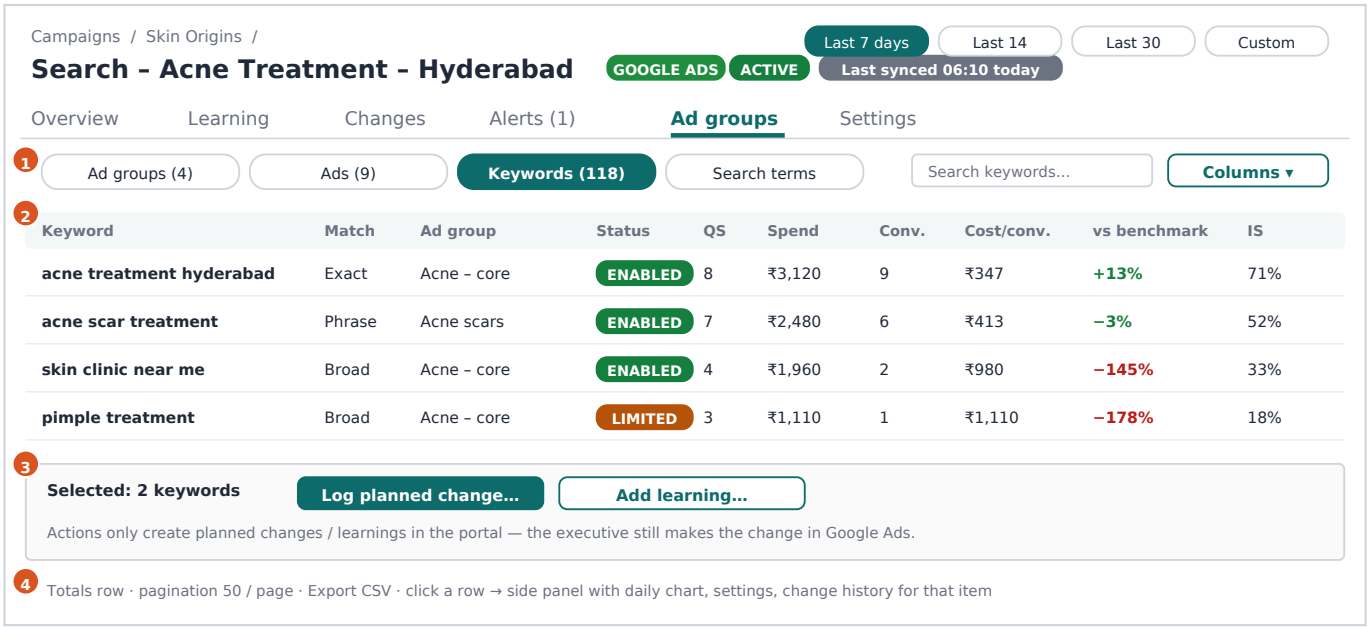


Figure 8 — Ad groups tab, Google Search campaign, Keywords sub-tab.

Platform / level	Columns (default; more via “Columns”)	Row click / extras
Meta — Ad sets	Name, status, learning stage, budget (if ABO), audience summary, placements, spend, results, cost / result, vs benchmark, CTR, frequency, CPM	Side panel: daily chart, targeting detail, change history. Expand → its ads
Meta — Ads	Thumbnail, name, format (reel / image / carousel), status / review, creative age (days), spend, results, cost / result, CTR, hook rate, hold rate, quality / engagement / conversion ranking	Preview of the creative; fatigue flag when age > 21 days and CTR falling
Google — Ad groups	Name, status, spend, conversions, cost / conv., vs benchmark, CTR, avg. CPC, impression share	Expand → ads and keywords of that group
Google — Ads	Headline preview, type (RSA etc.), ad strength, status / policy, spend, conversions, CTR, cost / conv.	Asset performance labels (Best / Good / Low)
Google — Keywords	Keyword, match type, ad group, status, Quality Score, spend, conv., cost / conv., vs benchmark, impression share, top-of-page rate	Figure 8

Platform / level	Columns (default; more via “Columns”)	Row click / extras
Google — Search terms	Search term, matched keyword, match type, spend, clicks, conv., cost / conv., added / excluded	Select → “Log planned change: add as negative / add as keyword”
Google — Asset groups (PMax)	Name, status, ad strength, spend, conv., conv. value, ROAS, channel split	Expand → assets with performance labels; audience signals

“vs benchmark” in these tables compares each row's cost per result with the **campaign's** benchmark, so the team can see which ad sets, keywords or ads pull the campaign away from it.

8B.3 Alerts tab

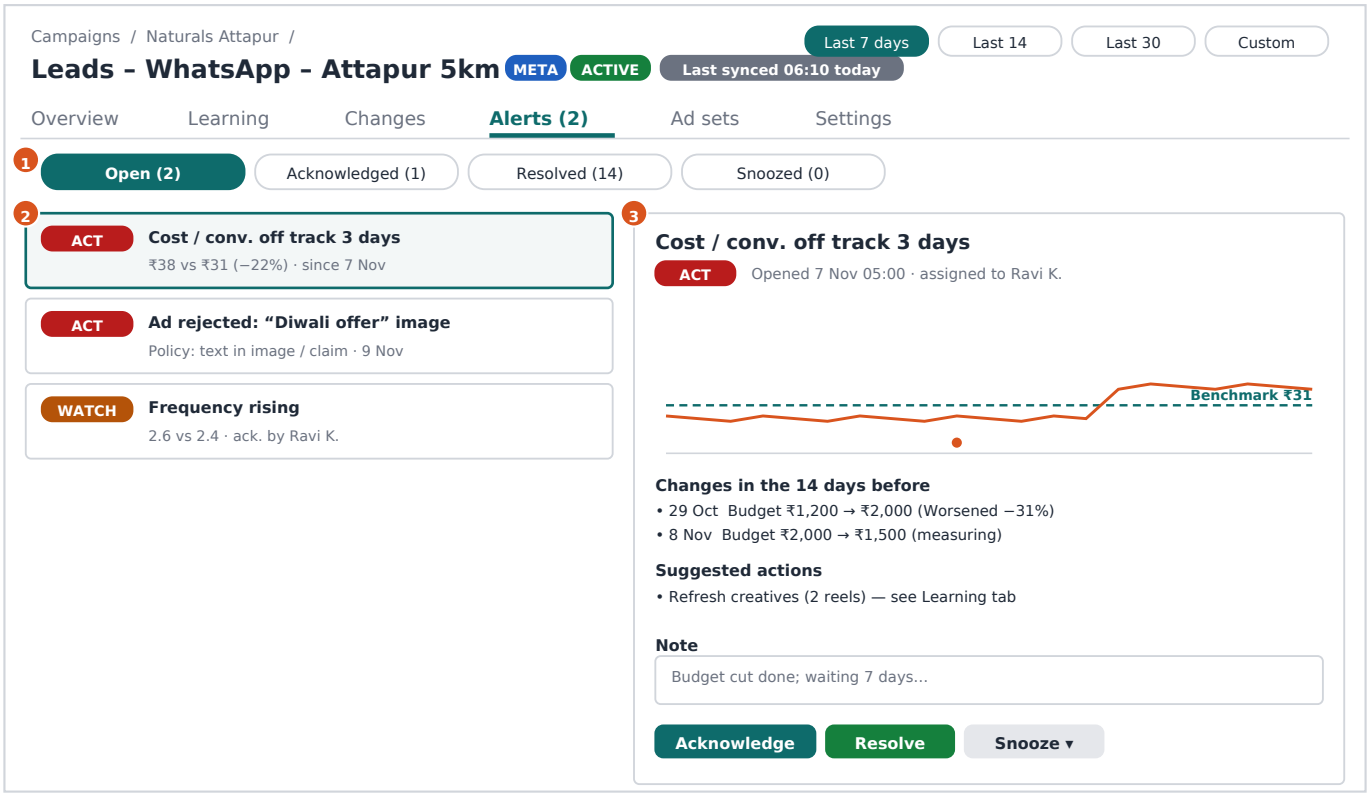


Figure 9 — Alerts tab: list (left) and selected alert detail (right).

- **Alert types shown:** benchmark-gap alerts (Act / Watch, section 7.5) and hard-rule alerts. Meta hard rules: zero delivery 24 h, ad rejected, ad set learning limited, account disabled / payment failed, spend cap near, pixel / CAPI events stopped. Google hard rules: zero impressions 24 h, ad disapproved, limited by budget, billing issue, conversions = 0 while clicks continue.
- **Detail panel:** KPI chart around the alert with benchmark line and change markers; changes in the 14 days before; suggested actions (links to Learning tab recommendations); notes thread; Acknowledge · Resolve (note required) · Snooze (1 / 3 / 7 days, note required).
- **States:** Open → Acknowledged → Resolved; auto-resolves when the KPI is back to On track for 3 days (shown as “Auto-resolved”).
- Tab label shows the open count. Resolved alerts keep their history and appear as markers on the Overview / Learning charts.

8B.4 Settings tab

Portal-side settings for how the campaign is measured, alerted on and owned. **Nothing here changes the campaign on Meta or Google** — platform settings are shown read-only on Overview ⑧.

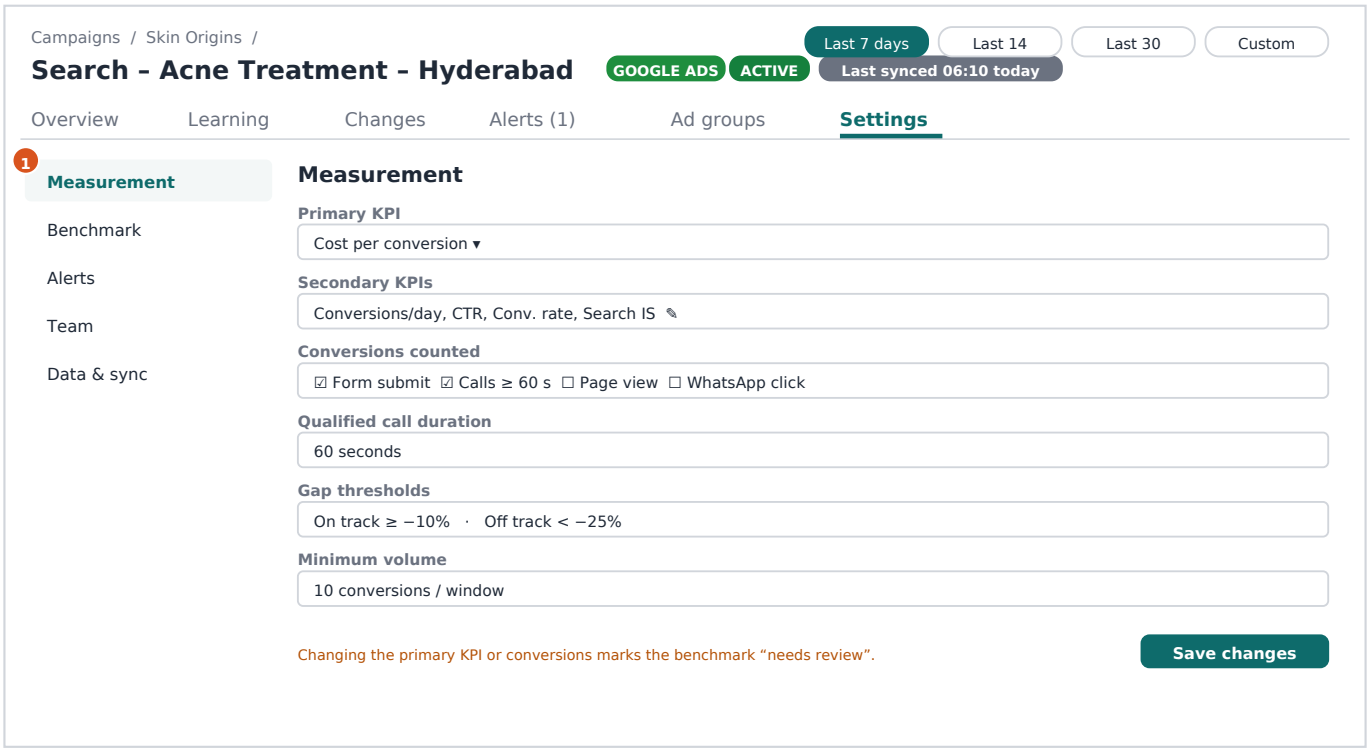


Figure 10 — Settings tab (Google example), Measurement section selected.

Section	Fields	Meta Ads	Google Ads
Measurement	Primary KPI; secondary KPIs (up to 4); what counts as a result; gap thresholds; minimum volume	Result = one action type (e.g. messaging conversation started, lead, purchase); attribution window shown read-only	Conversions counted = pick conversion actions; qualified-call duration (default 60 s)
Benchmark	Default method, window length, look-back, auto-advance, suggest updates, approval required; link to History	Same	Same
Alerts	Which alert rules are on for this campaign; Watch / Act sensitivity override; who gets alerts (executive, team lead, AM); channels (portal / email / WhatsApp); quiet hours	Meta hard rules list	Google hard rules list
Team	Assigned paid marketing executive, account manager, team lead; client contact for client-target benchmarks	—	—
Data & sync	Connection status, last synced, last error, ad account / customer ID (read-only), currency, **Re-sync now** button (admin), Exclude from portfolio insights, Archive in portal	Ad account act_…	Customer ID 123-456-7890

Permissions: executives can edit Alerts (own notifications) only; team lead / account manager / admin can edit Measurement, Benchmark and Team (section 13).

9. Data model (Supabase / Postgres)

New tables for this spec. The shared tables from the roadmap (`ad_accounts`, `ad_entities`, `ad_metrics_daily`, `campaign_changes`, `change_annotations`, `planned_changes`, `alerts`, `change_impact`) are assumed. All tables: `id uuid pk default gen_random_uuid()`, `created_at`, `updated_at`, RLS on.

```
campaign_kpi_config
  campaign_id uuid fk ad_entities  (unique)   primary_kpi text      primary_direction text  -- 'lower'|'higher'
  secondary_kpis text[]                       min_results int default 10
  qualified_call_seconds int default 60       last_goal_change_at timestamptz

campaign_benchmarks
  campaign_id uuid fk                         platform text          -- 'meta'|'google'
  mode text            -- 'auto'|'manual'|'client_target'
  method text          -- 'top3'|'best'|'p80'|null
  window_days int      lookback_days int      min_results int
  values jsonb         -- {"cost_per_result": 31.0, "results_per_day": 19, "ctr_link": 0.021, ...}
  locked_kpis text[]   edited_kpis text[]
  best_windows daterange[]                    eligible_windows int   excluded jsonb  -- {"learning":2,"low_volume":1,...}
  attribution_setting text                    currency text
  reason text          agreed_with text       agreed_on date
  status text          -- 'active'|'pending'|'suggested'|'superseded'|'rejected'
  auto_advance bool default false             suggest_updates bool default true
  created_by uuid      approved_by uuid       approved_at timestamptz   superseded_at timestamptz
  -- partial unique index: one 'active' row per campaign_id

campaign_settings_snapshots              -- daily, for "What was different in the best period"
  campaign_id uuid     date date             settings jsonb  -- budget, bid strategy, targets, audience summary,
                                                              -- creatives [{id,type,age_days}], placements, keywords summary
  unique (campaign_id, date)

campaign_learnings
  campaign_id uuid     tag text   -- 'worked'|'didnt_work'|'hypothesis'|'watch'
  text text            source text -- 'impact'|'ai'|'person'
  confidence text      -- 'high'|'medium'|'low'
  evidence jsonb       change_ids uuid[]      pinned bool default false
  status text          -- 'active'|'dismissed'
  dismissed_reason text                       created_by uuid

ai_recommendations                          (from roadmap; add)
  campaign_id uuid     title text   detail text   confidence text   evidence jsonb
  status text          -- 'open'|'accepted'|'modified'|'rejected'|'expired'
  decided_by uuid      decided_at timestamptz   reject_reason text
  planned_change_id uuid                       outcome_verdict text

campaign_notes
  campaign_id uuid     author uuid   body text   mentions uuid[]   parent_id uuid null

learning_state (view)  -- per campaign: 'collecting' | 'tracking', new_best bool, days_of_data, results_in_lookback
```

10. API

Implement as Supabase Edge Functions (or RPC) behind the portal's existing auth. All responses JSON; dates ISO-8601; money as numbers in account currency.

Method & path	Purpose
`GET /campaigns?client&platform&status&gap&mode`	Campaign list with primary KPI, benchmark, gap, status, alerts count
`GET /campaigns/:id/learning?range=7d`	Everything the Learning tab needs in one call (see example)
`GET /campaigns/:id/timeline?kpi&from&to`	Daily series + benchmark + best windows + platform-learning bands + change markers
`GET /campaigns/:id/best-period-diff`	Settings in best windows vs today
`POST /campaigns/:id/benchmarks/preview`	Body: mode, method, window_days, lookback_days, min_results → computed values (no save)
`POST /campaigns/:id/benchmarks`	Save a benchmark (auto values, manual values, locks, reason, options)
`POST /benchmarks/:id/approve | /reject`	Team lead / admin
`GET /campaigns/:id/benchmarks`	History
`POST /campaigns/:id/new-best/dismiss`	Hide new-best flag for 14 days
`GET /campaigns/:id/learnings?tag&page`	Learnings feed
`POST /campaigns/:id/learnings · PATCH /learnings/:id`	Add / edit / pin / dismiss

Method & path	Purpose
`GET /campaigns/:id/recommendations`	Open recommendations
`POST /recommendations/:id/accept | /modify | /reject`	Accept / Modify create a planned change
`GET|POST /campaigns/:id/notes`	Team notes
`PATCH /campaigns/:id/kpi-config`	Change primary / secondary KPIs (flags benchmark for review)

Example — GET /campaigns/:id/learning?range=7d


```
{
  "campaign": {"id": "…", "name": "Leads – WhatsApp – Attapur 5km", "platform": "meta", "status": "ACTIVE",
               "client": "Naturals Attapur", "currency": "INR", "last_synced_at": "2026-11-10T00:40:00Z",
               "in_platform_learning": false},
  "learning_mode": {"state": "tracking", "new_best": false, "days_of_data": 94},
  "benchmark": {"id": "…", "mode": "auto", "method": "top3", "status": "active",
                "best_windows": [["2026-10-12","2026-10-18"],["2026-09-21","2026-09-27"],["2026-09-03","2026-09-09"]],
                "approved_by": "Ravi K.", "approved_at": "2026-10-20T05:12:00Z", "needs_review": false},
  "kpis": [
    {"key": "cost_per_result", "label": "Cost / conversation", "primary": true, "direction": "lower",
     "current": 38.2, "benchmark": 31.0, "gap": -0.232, "status": "off_track", "volume_ok": true,
     "sparkline": [33.1, 35.0, 36.4, 39.9, 37.2, 40.1, 38.2]},
    {"key": "results_per_day", "label": "Conversations / day", "current": 14, "benchmark": 19, "gap": -0.263,
     "status": "off_track", "volume_ok": true, "sparkline": [17,16,15,13,14,12,14]}
  ],
  "open_recommendations": 2, "unread_learnings": 1, "open_alerts": 2
}
```


11. Background jobs

Job	Schedule (IST)	Does	Notes
Metrics sync	Daily 02:00	Meta + Google daily metrics; re-pull last 28 / 30 days	Roadmap Phase 1
Settings snapshot	Daily 02:30	Write `campaign_settings_snapshots` for every active campaign	Needed for best-period diff
Change capture	Daily 03:00	Meta activity log + Google change_event → `campaign_changes`; verify planned changes	Roadmap Phase 2
Gap & alerts	Daily 05:00	Recompute gap/status per campaign; raise / clear alerts; new-best detection	After sync completes
Auto benchmark	Weekly Mon 04:00, and on demand	Recalculate auto benchmarks; create suggested rows; auto-advance where enabled	Also run when a campaign first reaches 21 days
Change impact	Daily 05:30	Compute verdicts for changes whose after-window has completed → auto learnings	Roadmap section 5
AI learnings & recommendations	Weekly Mon 06:00, and “Refresh” button (max 3/day/campaign)	Generate learnings + up to 3 recommendations per active campaign	Section 12
Daily digest	Daily 09:30	Per executive: Act alerts, Watch list, changes to annotate, new recommendations	In-portal + chosen channel

- Use `pg_cron` + Edge Functions (same pattern as the existing Instagram sync). Every job writes a row to `job_runs` (job, started, finished, status, error) — the portal's connection-health and “last synced” come from here.
- Jobs are idempotent: re-running a day overwrites, never duplicates.

12. AI learnings contract

The AI never reads raw API data and never invents numbers. It receives a structured bundle and must return JSON matching the schema; anything that fails validation is dropped.

Input bundle (per campaign)

```
{ campaign: {platform, goal, primary_kpi, currency, client_vertical},
  benchmark: {mode, values, best_windows},
  daily_metrics: last 90 days,          settings_now, settings_best_period,
  changes: last 60 days with annotations and impact verdicts,
  existing_learnings: active ones (to avoid duplicates),   team_notes: last 30 days,
  portfolio_patterns: similar changes on other campaigns of the same platform+goal with verdicts (anonymised) }
```

Output schema

```
{ "learnings": [ { "tag": "worked|didnt_work|hypothesis|watch", "text": "≤ 160 chars",
                   "confidence": "high|medium|low", "change_ids": ["…"], "evidence": ["metric statements with numbers from
                       input"] } ],
  "recommendations": [ { "title": "≤ 80 chars", "detail": "≤ 300 chars", "confidence": "high|medium|low",
                         "expected_effect": {"kpi": "cost_per_result", "direction": "down", "size_pct": 15},
                         "evidence": ["…"], "similar_cases": {"worked": 5, "total": 6} } ] }
```

- Max 5 learnings and 3 recommendations per run. Confidence **high** only with ≥ 5 similar cases or a direct impact verdict.
- Every number in *evidence* must exist in the input bundle — validate by matching; drop the item otherwise.
- Recommendations are suggestions only. Nothing is ever written to Meta or Google.
- Store prompt version, model and input hash with each output for debugging.

13. Roles and permissions

Action	Executive	Team lead	Account mgr	Admin
View Learning tab	✓ (assigned clients)	✓	✓	✓
Save benchmark	✓ → pending if approval required	✓ (active)	✓ (client target)	✓
Approve / reject benchmark	—	✓	—	✓
Accept / modify / reject recommendation	✓	✓	—	✓
Add / edit own learnings, notes	✓	✓	✓	✓
Dismiss auto / AI learnings	✓	✓	—	✓
Edit KPI config, thresholds	—	✓	✓	✓
Delete benchmark history	—	—	—	✓ (soft delete)

Admin setting: **“Executives' benchmarks need team-lead approval”** (default ON). Enforce with Supabase RLS, not only in the UI.

14. Acceptance criteria and test cases

14.1 Acceptance criteria

1. Every active Meta and Google campaign has a Learning tab reachable at `/campaigns/:id/learning`.
2. Campaigns with < 21 days or no eligible window show **Collecting data** with correct progress, and allow a manual benchmark.
3. Auto benchmark values match the algorithm in 7.2 for the test fixtures below (to 2 decimal places).
4. Manual and client-target benchmarks save with a required reason; previous benchmark becomes *superseded*; history shows all versions.
5. With approval required, an executive's benchmark is *pending* and the previous one stays active until approved.
6. KPI cards show current, benchmark, signed gap and status following 7.5; “not enough data” and “in platform learning” states render correctly.
7. Chart shows benchmark line, shaded best windows, platform-learning bands and change markers with working tooltips.

8. Best-period diff shows highlighted differing rows for both platforms.
9. New best appears when the rule in section 6 is met, and dismiss hides it for 14 days.
10. Accepting a recommendation creates a planned change that appears in the Changes inbox and is verified the next day.
11. Stale-data banner appears when last sync is > 24 h old.
12. All of the above work for both a Meta and a Google campaign of each primary-KPI type in 7.1.

14.2 Benchmark test fixtures

#	Scenario	Expected
T1	90 days of steady data, 3 clearly best non-overlapping weeks	Top-3 picks those 3 weeks; values from combined totals
T2	Best week overlaps a Meta learning phase by 4 days	That week excluded; reason counted under “learning”
T3	A week with CPL ₹5 but only 4 leads	Excluded (low volume)
T4	A week with half-normal spend and great CPL	Excluded (low spend)
T5	Objective changed 20 days ago	Only windows after the change; state = Collecting data if < 1 eligible window
T6	Day with 0 conversions and 4× median clicks	Windows containing it excluded (tracking issue)
T7	Only 2 eligible non-overlapping windows	Uses 2; strip says “based on 2 windows”
T8	ROAS campaign (higher is better)	Best = highest ROAS; gap sign correct
T9	Locked CPL KPI, weekly recalculation produces lower CPL	CPL unchanged, other KPIs updated, suggested row created
T10	Manual benchmark, campaign 45% worse for 28 days	Soft “review this target?” prompt shown; no automatic change

15. Milestones

#	Milestone	Includes	Depends on
M1	Data ready	Metrics sync both platforms, settings snapshots, job_runs, stale-data banner	Roadmap Phase 1
M2	Benchmark core	KPI config, auto + manual + client-target benchmarks, drawer, history, approval, learning-mode states	M1
M3	Learning tab v1	Strip, KPI cards, chart (benchmark line + best windows), campaign list columns	M2
M4	Changes + impact	Change capture, Changes inbox, chart change markers, best-period diff, auto learnings from impact	M1
M5	Alerts on benchmark	Gap-based alerts, alerts centre and tab, daily digest	M3
M6	AI	AI learnings and recommendations, accept → planned change → verification	M4 + 2–3 months of annotated changes for good quality

Open questions for the product owner

- Default window length 7 days for all clients, or 14 days for low-volume clients (e.g. under 30 results a month)?
- Should team-lead approval be required for executives' benchmarks from day one? (Spec default: yes.)
- Which channel for the daily digest — portal only, email or WhatsApp?
- Minimum qualified-call duration per client for Google call campaigns (default 60 s)?
- Will clients ever see the benchmark? If yes, a client-facing read-only view needs its own design pass.